Nagar Yuwak Shikshan Sanstha's

Yeshwantrao Chavan College of Engineering

(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)
Hingna Road, Wanadongri, Nagpur - 441 110
NAAC A++
Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu

Department of Artificial Intelligence and Data Science

Vision of the Department

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission of the Department

To nurture computer engineering professionals by offering:

- *Dynamic learning environment that encourages problem-solving using emerging technology*
- *National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry–institute collaboration*
- *Holistic, ethical and sustainable development for lifelong success.*

Session 2026-2027

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

(Signature and Date in Handwritten)

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Session	**2026-27 (Odd)**	**Course Name**	**LAB: Optimization technique**
Semester	**7th**	**Course Code**	**23ADS1708**
Roll No	39	**Name of Student**	**Hrutik Gujarkar**

Practical Number	**7**
Course Outcome	CO1: Understand unconstrained and constrained optimization techniques. CO2: Apply non-linear optimization techniques to solve engineering problems. CO3: Analyze optimization problems using operations research techniques. CO4: Evaluate transportation and transshipment optimization problems.
Aim	To determine the best-fit curve using the Nonlinear Least Squares Method.
Problem Definition	To fit the exponential model $y = a \cdot e^{bx}$ to the given data points (x = 0, 1, 2, 3, 4, 5 and y = 2.0, 3.4, 5.3, 9.1, 14.5, 24.6) by minimizing the sum of squared residuals, and to determine the parameters a and b of the best-fit curve.
Theory (100 words)	The Nonlinear Least Squares method fits a model $y = f(x; \beta)$ that is nonlinear in its parameters β observed data (x_i, y_i). The best parameters minimize the sum of squared residuals $S(\beta) = \Sigma [y_i - f(x$ $\beta)]^2$. Unlike linear least squares, there is no closed-form solution, so iteration is required. Th Gauss-Newton method linearizes the model using its Jacobian J and updates $\Delta\beta = (J^T J)^{-1} J^T r$, whe r is the residual vector. The Levenberg–Marquardt algorithm, used by SciPy's curve_fit, adds damping term for stability. The fit is judged using SSE and R^2.
	Step-by-Step Method 1. Select the model $y = a \cdot e^{bx}$ and enter the observed data. 2. Define the residual of each point as the observed y minus the model value $a \cdot e^{bx}$. 3. Define the objective $S(a, b) = \Sigma r_i^2$. 4. Choose an initial guess for the parameters a and b.

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	5. Compute the Jacobian of the model with respect to a and b at the current guess. 6. Update the parameters using the Gauss-Newton step (with damping in Levenberg–Marquardt). 7. Repeat until the sum of squared residuals stops decreasing. 8. Calculate SSE and R^2 to check the quality of the fit.
Procedure and Execution (100 Words)	Algorithm: 1. Enter the data points (x, y). 2. Define the nonlinear model $y = a \cdot e^{bx}$. 3. Provide an initial guess for a and b. 4. Fit the model using nonlinear least squares (curve_fit). 5. Obtain the estimated parameters a and b. 6. Calculate the fitted values and residuals. 7. Calculate SSE and R^2. 8. Display the parameters and the best-fit curve.

	Code:
	`import numpy as np`
	`from scipy.optimize import curve_fit`
	
	`# Observed data`
	`x = np.array([0, 1, 2, 3, 4, 5], dtype=float)`
	`y = np.array([2.0, 3.4, 5.3, 9.1, 14.5, 24.6])`
	
	`# Model to be fitted: y = a * exp(b * x)`
	`def model(x, a, b):`
	`  return a * np.exp(b * x)`
	
	`# Nonlinear least squares fit (initial guess a = 1, b = 0.1)`
	`params, cov = curve_fit(model, x, y, p0=[1.0, 0.1])`
	`a, b = params`
	
	`# Fitted values and residuals`
	`y_fit = model(x, a, b)`
	`residuals = y - y_fit`
	`sse = np.sum(residuals**2)`
	`sst = np.sum((y - np.mean(y))**2)`
	`r2 = 1 - sse/sst`
	
	`print("Model: y = a * exp(b * x)")`
	`print("\nx  y(actual)  y(fitted)  residual")`
	`for xi, yi, fi, ri in zip(x, y, y_fit, residuals):`
	`  print(f"{xi:.0f}  {yi:8.3f}  {fi:8.3f}  {ri:8.3f}")`
	
	`print("\nEstimated Parameters:")`
	`print("a = %.4f" % a)`
	`print("b = %.4f" % b)`
	`print("\nSum of Squared Errors (SSE) = %.4f" % sse)`
	`print("R-squared = %.4f" % r2)`
	`print("\nBest-fit curve: y = %.4f * exp(%.4f * x)" % (a, b))`

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	Output:
	`Model: y = a * exp(b * x)`
	
	`x  y(actual)  y(fitted)  residual`
	`0  2.000  1.960  0.040`
	`1  3.400  3.249  0.151`
	`2  5.300  5.384  -0.084`
	`3  9.100  8.921  0.179`
	`4  14.500  14.784  -0.284`
	`5  24.600  24.499  0.101`
	
	`Estimated Parameters:`
	`a = 1.9604`
	`b = 0.5051`
	
	`Sum of Squared Errors (SSE) = 0.1542`
	`R-squared = 0.9996`
	
	`Best-fit curve: y = 1.9604 * exp(0.5051 * x)`

Output Analysis	The program fitted the model $y = a \cdot e^{bx}$ to the data and obtained a = 1.9604 and b = 0.5051. Th residuals are small, with a maximum absolute value of 0.284. The sum of squared errors is 0.154 and $R^2 = 0.9996$, which shows that the exponential curve describes the data very well. The positiv value of b indicates exponential growth.
Link of student Github profile where lab assignment has been uploaded	https://github.com/HrutikGujarkar/Optimizaton-techniques-Practicals
Conclusion	The best-fit curve was successfully determined using the **Nonlinear Least Squares Method**. Th best-fit curve is $\mathbf{y = 1.9604 \cdot e^{0.5051x}}$ with $R^2 = 0.9996$.

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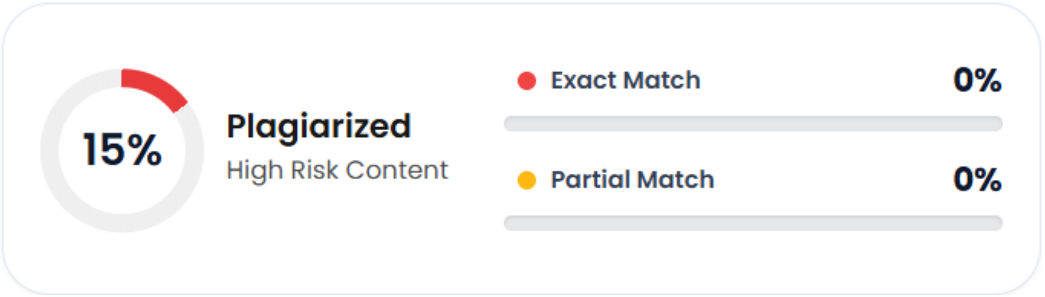
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